

Abhay Manhas

♥ Pathankot, Punjab | ✉ manhasabhay000@gmail.com | 📞 +91 8195847151 | 🌐 abhaymanhas19

PROFILE

- Engineer robust backend systems with Python, Django, and Django REST Framework, leveraging PostgreSQL for data persistence and WebSockets for real-time communication in high-performance applications.
- Integrate diverse AI APIs (OpenAI, Gemini, Azure AI, Deepgram, Assembly.ai) for multimodal intelligence solutions.
- Architect AI-driven solutions including RAG chatbots and NLP pipelines integrated with speech-to-text APIs.
- Orchestrate event-driven microservices using RabbitMQ and Kubernetes for resilient cloud deployments.

PROFESSIONAL EXPERIENCE

Senior Python Developer, *Shubpy Pvt. Ltd., Chandigarh*

Mar 2023 – Present

- Architected a hybrid-cloud AI orchestration pipeline using OpenRouter to dynamically route jobs to Gemini or Azure OpenAI, **reducing AI task costs by 35%**.
- Developed an advanced RAG-based AI chatbot for document querying, achieving a **contextual accuracy score of over 0.85** using semantic search.
- Built a MapReduce-style analysis service with Celery to process large documents, enabling summarization of 500+ page texts in **under 1 minute**.
- Engineered a real-time progress-tracking interface using Django Channels and WebSockets, reducing **support queries by 60%**.
- implemented the company's first CI/CD pipeline in Azure DevOps, achieving unit test coverage and reducing deployment incidents by **90%**.
- Applied exponential backoff with jitter and a Redis-based distributed semaphore for global rate limiting, ensuring **100% API compliance and 75% fewer timeouts**.
- Designed message queue system with **RabbitMQ** ensuring fault-tolerant communication between services.

Python Developer, *CNT Technologies, Chandigarh*

June 2022 - Dec 2022

- Built a real-time chat application using asynchronous programming and WebSockets, enabling instant communication for **100+ concurrent users with message latency under 100ms**.
- Optimized PostgreSQL queries by **70%**, reducing API latency from **1200ms to 350ms** for high-traffic endpoints.
- Implemented event-driven microservices using **RabbitMQ** for asynchronous task processing.
- Mentored **3 junior developers** in Python best practices during feature implementation, **ensuring 100% sprint goal completion**.

SKILLS

Languages Frameworks: Python, Django, Django REST Framework (DRF), Channels, WebSockets, Redis, Asyncio, Celery, HTTPX, PyTorch(learning), OpenCV.

DevOps Cloud Infrastructure: Azure Kubernetes Service (AKS), Azure DevOps (CI/CD), Docker, Azure Functions, Blob Storage, Azure Database for PostgreSQL.

Architecture Design Patterns Microservices, Event-Driven Architecture (RabbitMQ), Retrieval Augmented Generation (RAG), Semantic Search, MapReduce Patterns.

AI/ML API Integrations: Gemini, OpenAI, OpenRouter, Grok, Claude, Deepgram, AssemblyAI, DeepL, LiveKit, Tavus, RAG, NLP.

Observability Monitoring: Prometheus, Grafana, Flower.

Testing Quality : Pytest, unittest, Mocking, End-to-End Test Strategy.

EDUCATION

Bachelor Of Computer Application

06/2019 – 06/2022

Gura Nanak dev university, Amritsar.

Senior Secondary Education

03/2018 – 03/2019

Govt. High Secondary School , Bhoa, Pathankot